10163 - Revealing Unexpected Carbon Chemistry in the Oxygen-Rich Ejecta of a Dying Star

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
ON-SOUCRE OBSERVATIONS (NGC6302 NIRSPEC)				
	2	NGC 6302 G235H/F170LP	NIRSpec IFU Spectroscopy	(1) NGC-6302
	3	NGC 6302 G395H/F290LP	NIRSpec IFU Spectroscopy	(1) NGC-6302
BACKGROUND OBSERVATIONS (NGC6302_NIRSPEC)				
	6	Background G395H/F290LP	NIRSpec IFU Spectroscopy	(2) NGC-6302-BACKGROUND
	5	Background G235H/F170LP	NIRSpec IFU Spectroscopy	(2) NGC-6302-BACKGROUND

ABSTRACT

Archival JWST/MIRI observations of the oxygen-rich planetary nebula NGC 6302 - whose central star was too massive to become carbon-rich - have revealed three chemical surprises that point to previously unknown pathways to carbonaceous molecular species in this environment: abundant CH3+, a key driver of organic chemistry in star-forming regions but unexpected in evolved stars; significant PAH spectral variations indicating in-situ processing; and CO2 ice in a dusty torus. Characterizing these novel chemical pathways will address a longstanding puzzle: how can carbon-

bearing molecules become abundant in oxygen-rich planetary nebulae?

We propose NIRSpec-IFU observations (1.66–5.27 micron, R~2700) to map the chemical pathways and physical conditions driving this rich carbon chemistry. Our objectives are to: (1) determine whether CH_3^+ forms via UV-pumped H_2 chemistry as in star-forming regions by measuring H_2 excitation and searching for CH^+; (2) characterize PAH size distributions and aliphatic/aromatic content from 3.3/3.4 micron band ratios; (3) determine ice compositions (H_2O, CO, CO_2, and other species) and evaluate ice-gas chemistry potential; and (4) constrain physical conditions using H_2 excitation diagrams, atomic fine-structure lines, and overlapping MIRI data.

This program will reveal chemical pathways in organic astrochemistry influenced by UV irradiation and shocks, determining what carbonaceous material low- and intermediate-mass stars inject into the ISM. Our findings will apply to carbon chemistry in star-forming regions, protoplanetary disks, starburst galaxies, and AGN environments.

OBSERVING DESCRIPTION

This program includes a 5x2 tile mosaic with NIRSpec-IFU with the disperser-filter combinations G235H/F170LP and G395H/F290LP using the NRSIRS2RAPID readout mode and a 4-point extended source dither pattern. Background observations and leakcal exposures are also included, in a non-interruptible sequence. The orientation of the mosaic is optimized to cover all the Regions of Interest for our science case, which constrains the aperture PA range.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-6302	RA: 17 13 44.5030 (258.4354292d) Dec: -37 06 12.13 (-37.10337d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. *Category=ISM* *Description=[Planetary nebulae]* *Extended=YES*				
(2)	NGC-6302-BACKGROUND	RA: 17 13 30.6600 (258.3777500d) Dec: -37 04 23.40 (-37.07317d) Equinox: J2000	Epoch of Position: 2000	
Comments: *Category=ISM* *Description=[Planetary nebulae]* *Extended=YES*				

Observation

Proposal 10163, Observation 2: NGC 6302 G235H/F170LP Fri Mar 13 19:01:02 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[Background G235H/F170LP (Obs 5)]

Diagnostics

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-6302	RA: 17 13 44.5030 (258.4354292d) Dec: -37 06 12.13 (-37.10337d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	5	10.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235H/F170LP	NRSIRS2RAPID	5	4	false	true	NONE	4	16	1400.533	264019
2	G235H/F170LP	NRSIRS2RAPID	5	4	true	true	NONE	4	16	1400.533	264019

Special Requirements	
	Aperture PA Range 235.97164917 to 238.97164917 Degrees (V3 97.0 to 100.0) Sequence Observations 2, 5, Non-interruptible

Observation

Proposal 10163, Observation 3: NGC 6302 G395H/F290LP

Fri Mar 13 19:01:02 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[Background G395H/F290LP (Obs 6)]

Diagnostics

(Visit 3:1) Warning (Form): Data Excess over lower threshold

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-6302	RA: 17 13 44.5030 (258.4354292d) Dec: -37 06 12.13 (-37.10337d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	5	10.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2RAPID	5	4	false	true	NONE	4	16	1400.533	264019
2	G395H/F290LP	NRSIRS2RAPID	5	4	true	true	NONE	4	16	1400.533	264019

Special Requirements	
	Aperture PA Range 235.97164917 to 238.97164917 Degrees (V3 97.0 to 100.0) Sequence Observations 3, 6, Non-interruptible

Observation

Proposal 10163, Observation 6: Background G395H/F290LP

Fri Mar 13 19:01:02 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NGC 6302 G395H/F290LP (Obs 3)]

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	NGC-6302-BACKGROUND	RA: 17 13 30.6600 (258.3777500d) Dec: -37 04 23.40 (-37.07317d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2RAPID	5	4	false	true	NONE	4	16	1400.533	

Special Requirements

Sequence Observations 3, 6, Non-interruptible

Observation

Proposal 10163, Observation 5: Background G235H/F170LP

Fri Mar 13 19:01:02 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NGC 6302 G235H/F170LP (Obs 2)]

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	NGC-6302-BACKGROUND	RA: 17 13 30.6600 (258.3777500d) Dec: -37 04 23.40 (-37.07317d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G235H/F170LP	NRSIRS2RAPID	5	4	false	true	NONE	4	16	1400.533	

Special Requirements

Sequence Observations 2, 5, Non-interruptible